Salima Lamsiyah

Teaching and Supervision Activities

Teaching Experience

- **Sept 2017–June 2022: Lycée Mathematics Teacher, Ministry of National Education, Morocco.** During the preparation of my Ph.D., I also served as a mathematics teacher at lycée level, as a civil servant employed by the Ministry of National Education. Over five academic years (**September 2017–June 2022**), I gained solid experience in pedagogy and classroom management. This role enabled me to develop key skills in teaching, explaining complex concepts in an accessible way, adapting to diverse learning styles, and transferring knowledge effectively. I also refined my abilities in communication, mentoring, and supporting students in their learning paths, which today contribute significantly to my work in higher education and research supervision.

- **July 2022–Present: Lecturer – Natural Language Processing, Department of Computer Science, FSTM, University of Luxembourg.** Taught the bachelor-level course *Natural Language Processing* (60 hours, theoretical and practical) for four consecutive academic years as the sole instructor.

  - 09/2025 – 01/2026: NLP, 5th Semester BICS (30 units)
  - 09/2024 – 01/2025: NLP, 5th Semester BICS (30 units)
  - 09/2023 – 01/2024: NLP, 5th Semester BICS (30 units)
  - 09/2022 – 01/2023: NLP, 5th Semester BICS (30 units)

  The course is designed as a progressive introduction to Natural Language Processing (NLP), beginning with fundamental concepts and methods, and advancing towards modern deep learning and generative approaches. It balances theoretical foundations with hands-on practice, ensuring students understand both classical techniques and the latest developments in the field. Teaching is complemented with regular assignments and project-based learning, enabling students to apply concepts in real-world scenarios. Final projects are renewed annually to reflect emerging trends and applications in NLP, encouraging creativity, critical thinking, and practical problem-solving skills.

- **February 2026: Invited Lecturer – Large Language Models, Lancaster University, United Kingdom.** Delivered invited teaching on Large Language Models (LLMs) for Bachelor's and Master's students.

- **2025–2026: Invited Lecturer at Summer Schools.** Delivered invited lectures at the *EUMaster4HPC Summer School 2026*, Luxembourg (July 2026); the summer school *"From Rules to Models"*, Alicante, Spain (June 2026); and the *RANLP 2025 Summer School "Deep Learning and LLMs for NLP"*, Varna, Bulgaria (September 2025).

- **09/2023 – 12/2023: Instructor – Introduction to AI and ChatGPT, Luxembourg Online Education Center A.s.b.l, in collaboration with the Luxembourg Ministry of Education (30 hours).** Designed and delivered a training course introducing participants to Artificial Intelligence and ChatGPT. The course covered fundamental AI concepts, Natural Language Processing, Generative

AI, prompting techniques for Large Language Models, and practical applications, including fine-tuning and Retrieval-Augmented Generation (RAG). The training received excellent feedback from participants.

- **2023–2024: Contributor – Elements of AI, University of Luxembourg.** Contributed to the delivery of the *Elements of AI* course to a broad public audience as part of the EOAI 2023 and EOAI 2024 initiatives.

Supervision of Master's Theses

- **July 2022–Present: Supervisor – Master's Students, Department of Computer Science, FSTM, University of Luxembourg.** Served as daily advisor for **eight Master's students** on research projects in Natural Language Processing, Explainable AI, Reinforcement Learning, Federated Learning, and Generative AI for Education. Responsibilities included continuous mentoring, guidance on research methodology, implementation, and scientific writing.
  - Supervised theses covering topics such as AI-generated text detection, generative AI for education, explainability in Transformer models, and federated approaches for NLP applications.
  - One of my students, **Keerthana Murugaraj**, received the **Best Master's Thesis Award (2023)** from the Faculty of Science, Technology and Medicine at the University of Luxembourg.

Supervision of Bachelor's Theses and Projects

- **July 2022–Present: Supervisor – Bachelor's Students, Department of Computer Science, FSTM, University of Luxembourg.** Supervised more than **40 Bachelor's students** across Bachelor's theses and semester projects in the fields of Natural Language Processing and Artificial Intelligence. Provided close guidance on project design, implementation, and reporting, with a strong focus on bridging theory and practice.
  - Projects covered a wide range of topics, including Generative AI, Computer Vision, NLP for Cybersecurity, NLP for Education, AI-based Recommender Systems, Multimodal Harmful Meme Detection, Conversational Agents, and Human-in-the-Loop Collaborative Storytelling.
  - Emphasis placed on developing students' critical thinking, problem-solving abilities, and scientific communication skills.